

The Collateral Network Concept: A Reassessment of the Anatomy of Spinal Cord Perfusion

OBJECTIVE Prevention of paraplegia following repair of thoracoabdominal aortic aneurysms (TAAA) requires understanding the anatomy and physiology of the blood supply to the spinal cord. Recent laboratory studies and clinical observations suggest that a robust collateral network must exist to explain preservation of spinal cord perfusion when segmental vessels are interrupted. An anatomical study was undertaken. **METHODS** Twelve juvenile Yorkshire pigs underwent aortic cannulation and infusion of a low-viscosity acrylic resin at physiological pressures. After curing of the resin and digestion of all organic tissue, the anatomy of the blood supply to the spinal cord was studied grossly and using light and electron microscopy. **RESULTS** All vascular structures $\geq 8\mu\text{m}$ in diameter were preserved. Thoracic and lumbar segmental arteries (SAs) give rise not only to the anterior spinal artery (ASA), but to an extensive paraspinous network feeding the erector spinae, iliopsoas, and associated muscles. The ASA, mean diameter $134\pm 20\mu\text{m}$, is connected at multiple points to repetitive circular epidural arteries with mean diameters of $150\pm 26\mu\text{m}$. The capacity of the paraspinous muscular network is 25-fold the capacity of the circular epidural arterial network and ASA combined. Extensive arterial collateralization is apparent between the intraspinal and paraspinous networks, and within each network. Only 75% of all SAs provide direct ASA-supplying branches. **CONCLUSIONS** The ASA is only one component of an extensive paraspinous and intraspinal collateral vascular network. This network provides an anatomic explanation of the physiological resiliency of spinal cord perfusion when SAs are sacrificed during TAAA repair.